

 YTU – Faculty of Chemical and Metallurgical Engineering, Questions and Answers Sheet	NOTE CHART					
	Q1	Q2	Q3	Q4	Q5	TOTAL
Student Name and Surname						
Students Number						
Course Name	Int. to Financial Mathematics (Online Midterm)		Exam Duration	90 min.	Exam Date	15.05.2020
Course Instructor Name and Surname	Assoc. Prof. Dr. Hale Gonce Kocken		Signature			
Student Disciplinary Regulations "and to make or attempt to make copies of exams to" the actual perpetrators are suspended from one or two semesters. (YÖK; 2547 Student Disciplinary Regulations, 9. Article)						

Q1) Consider an initial investment of \$200 using an amount function which is a second degree polynomial. You are given that the future values of this investment at $t = 2$ and $t = 3$ are 280 and 350, respectively. Determine the followings:

- The effective interest rate of the third year.
- The equivalent effective rate of discount between $t = 0.5$ and $t = 1.5$.
- The force of interest at time $t = 1.5$,
- The present value of 100 which is paid at time $t = 1.5$.

Q2) Aybars borrows 2000\$ at a nominal interest rate of 12% convertible quarterly. He repays the loan completely by making the following payments:

- 700\$ at the end of 4 months
- 500\$ at the end of 14 months
- R at the end of 18 months

Calculate R.

Q3) Teoman deposits an amount of money into a bank account. The interest will be credited at the following rates, respectively.

Investment period	Interest type	Interest rate
1. For the first two years	Simple interest	5%
2. Next 1 year 6 months	Nominal rate of discount convertible semiannually	8%
3. Next 2 year 3 months	Simple discount	10%
4. Next 1 year 3 months	Nominal rate of interest convertible quarterly	16%

For this seven-year financial investment, find the equivalent annual effective rate of interest.

Q4) The principal 1000 TL accumulates to 1800 TL at the end of 5 years. Find the corresponding nominal interest rate convertible quarterly.

Q5) The cash price of a house is \$100000. The buyer will make payments of \$1000 at the end of each month for ten years. The nominal rate of interest is 12% convertible monthly. Find the down payment that will be necessary.

I wish you success.